

Yuhan Chi

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Education

Fudan University

Shanghai, China

B.S. in Information and Computing Science & Artificial Intelligence (Double Degree) Sep. 2025 – Present

School of Mathematical Sciences

- GPA: 3.96/4.00 (most recent semester); Rank: 10th (top 5%) in cohort

Research Interests

- **Mathematical Foundations of Intelligence** — grounding AI trust in formal, mathematical terms rather than empirical benchmarks
- **Entrustable Sequential Decision-Making** — building agents that are mathematically verifiable and know when to act, defer, or bear consequences
- **Reinforcement Learning & World Models** — RL as a language for trust, uncertainty, and structured interaction with the environment

Selected Open-Source Projects

code-not-text

2026

github.com/chi-shan0707/code-not-text

- Paper-level research on cross-domain limits of hand-crafted CoT-surface features for predicting solution correctness from reasoning traces
- Evaluated on DeepSeek-R1-0528-Qwen3-8B across math, science, and coding; strong in math (AoA 0.958), below-chance in coding (AoA 0.434)
- Five independent checks (probes, feature sweep, MLPs, SSL, de-knotting) all converge to the same coding ceiling, ruling out capacity, engineering, scarcity, and noise — code correctness is not in the text
- Argued that CoT traces are not entrustable correctness instruments: they signal reasoning failure in math but only text fluency in coding, making execution feedback essential for verifiable agents
- Interactive demo at chi-shan0707.github.io/code-not-text/demo

TinyLoRA-GRPO-Coder

2026

github.com/Chi-Shan0707/TinyLoRA-GRPO-Coder

- Independent reimplement and adaptation of TinyLoRA + GRPO from *Learning to Reason in 13 Parameters*
- Migrated the method from math reasoning to verifiable competitive-programming code generation on Qwen2.5-Coder-3B
- Trained with a minimal number of shared trainable parameters; replaced static heuristics with real compile-and-run rewards
- Built the full pipeline end-to-end: data processing, training, multi-GPU setup, reward design, evaluation, and validation

microgpt.cpp

2026

github.com/Chi-Shan0707/microgpt.cpp

- Minimal GPT implementation from first principles in C++, motivated by understanding model internals without relying on high-level frameworks

Skills

- **Programming:** Python, C++, C, Java, Node.js, Lean 4
- **ML / AI:** PyTorch, deep learning, LLM fine-tuning, reinforcement learning, computer vision
- **Tools:** Git, GitHub, Linux, $\text{\LaTeX}$, Markdown, VS Code
- **Other:** Mathematical proof writing, technical writing, self-directed learning, competitive programming

Additional Information

- Actively self-studying mathematics, algorithms, and AI beyond formal coursework; maintains an active public GitHub profile documenting ongoing work
- Maintains github-unflag-playbook-cn, a Chinese playbook documenting GitHub account flagging, recovery cases, and appeal workflows. chi-shan0707.github.io/github-unflag-playbook-cn
- Contributor to FDUGuideBook/nav-site, a Fudan community site. github.com/FDUGuideBook/nav-site
- Personal website: chi-shan0707.github.io